

LAB 6

Dynamic Routing Configurations RIP, EIGRP, OSPF, BGP

OBJECTIVES

- To understand dynamic routing in computer networks.
- To configure and verify RIP, EIGRP, OSPF, and BGP.
- To observe automatic routing updates and path selection.
- To compare routing protocols based on performance and usage.

Software and Hardware Requirements

- Cisco Packet Tracer
- Windows Operating System
- Computer/Laptop
- Keyboard and Mouse

THEORY

Dynamic routing is a networking technique in which routers automatically discover, share, and update routing information using routing protocols. Unlike static routing, dynamic routing adapts to network changes such as link failures or topology updates without manual intervention. Dynamic routing protocols use algorithms and metrics to determine the best path for data transmission.

RIP (Routing Information Protocol)

RIP is a simple distance-vector routing protocol that uses hop count as its metric, with a maximum limit of 15 hops, making it suitable only for small networks. Routers exchange full routing tables every 30 seconds using the Bellman-Ford algorithm, which results in slow convergence and poor scalability. RIPv1 is classful and uses broadcast, while RIPv2 is classless, supports subnet masks, multicast updates, and authentication. Loop prevention mechanisms such as split horizon, route poisoning, and hold-down timers are used to reduce routing loops.

EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is an advanced distance-vector (hybrid) routing protocol developed by Cisco that uses a composite metric based on bandwidth and delay. It uses the Diffusing Update Algorithm (DUAL) to provide fast convergence and loop-free routing. EIGRP sends routing updates only when changes occur using partial and triggered updates, which reduces unnecessary network traffic. It is a classless routing protocol that supports VLSM, route summarization, authentication, and unequal-cost load balancing.

OSPF (Open Shortest Path First)

OSPF is a link-state routing protocol that uses cost as its metric, which is calculated based on bandwidth. Routers exchange link-state advertisements (LSAs) to build and maintain a complete view of the network topology and run the Dijkstra shortest path algorithm to determine the best path. OSPF supports classless routing, VLSM, and hierarchical network design using areas, which helps reduce routing overhead. It provides fast convergence and is widely used within large enterprise networks.

BGP (Border Gateway Protocol)

BGP is a path-vector routing protocol primarily used for routing between autonomous systems, especially on the Internet. Instead of using simple metrics, BGP makes routing decisions based on path attributes such as AS path, next hop, and routing policies. Routers exchange routing information using reliable TCP connections, ensuring stable communication. BGP is highly scalable and policy-based, making it suitable for large, complex inter-network environments.

Configuration for ALL:

Device	Interface / Subnet	IP Address	Subnet Mask	Default Gateway
Router0	FastEthernet0/0	192.168.1.1	255.255.255.0	N/A
Router0	FastEthernet0/1	10.0.0.1	255.0.0.0	N/A
Router1	FastEthernet0/0	192.168.2.1	255.255.255.0	N/A
Router1	FastEthernet0/1	10.0.0.2	255.0.0.0	N/A
PC1	Subnet 1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	Subnet 1	192.168.1.3	255.255.255.0	192.168.1.1
PC3	Subnet 2	192.168.2.2	255.255.255.0	192.168.2.1
PC4	Subnet 2	192.168.2.3	255.255.255.0	192.168.2.1

RIP (Routing Information Protocol)

CLI:

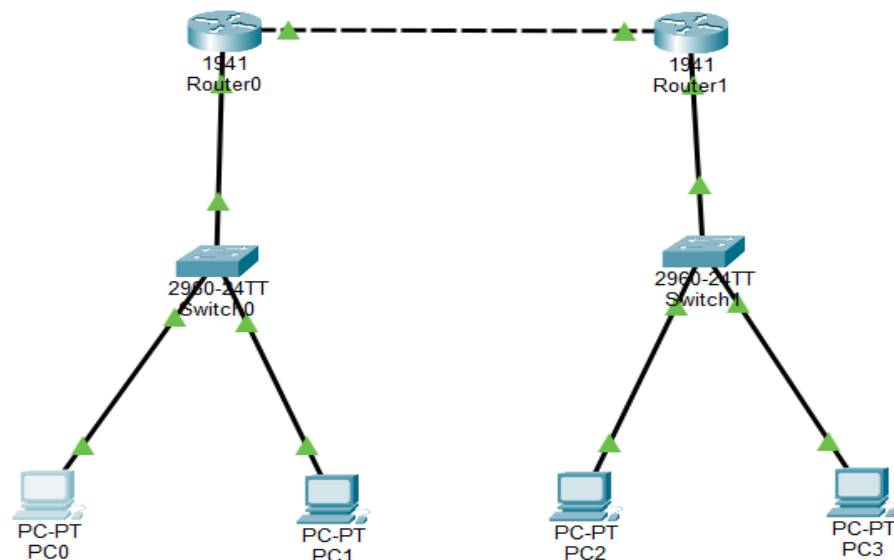
Router 0

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#
```

Router 1

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#
```

Network Topology



Ping Output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

EIGRP (Enhanced Interior Gateway Routing Protocol)

CLI:

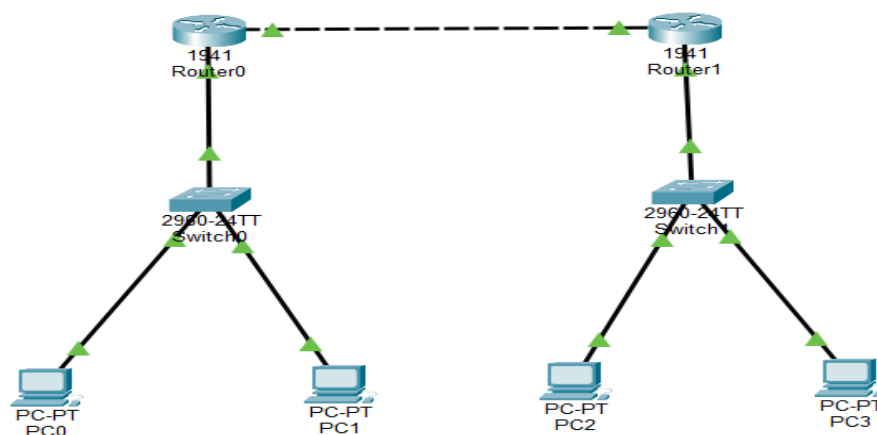
Router 0

```
Router(config)#router eigrp 100
Router(config-router)#network 198.168.1.0 0.0.0.255
Router(config-router)#network 10.0.0.0 0.0.0.255
Router(config-router)#no auto-summary
Router(config-router)#
```

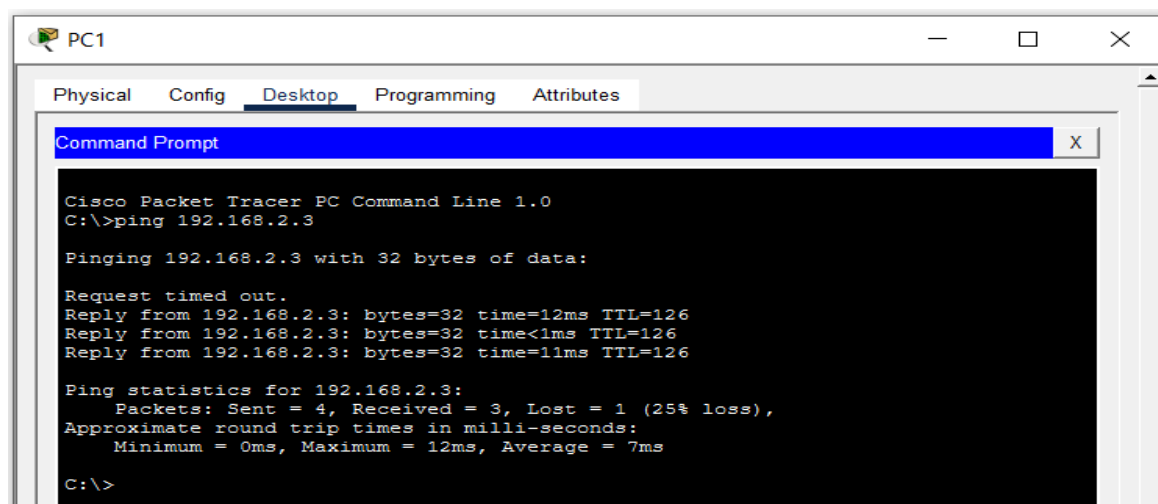
Router 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 100
Router(config-router)#network 192.168.2.0 0.0.0.255
Router(config-router)#network 10.0.0.0 0.0.0.255
Router(config-router)#no auto-summary
Router(config-router)#
```

Network Topology



Ping Output:



OSPF (Open Shortest Path First)

CLI:

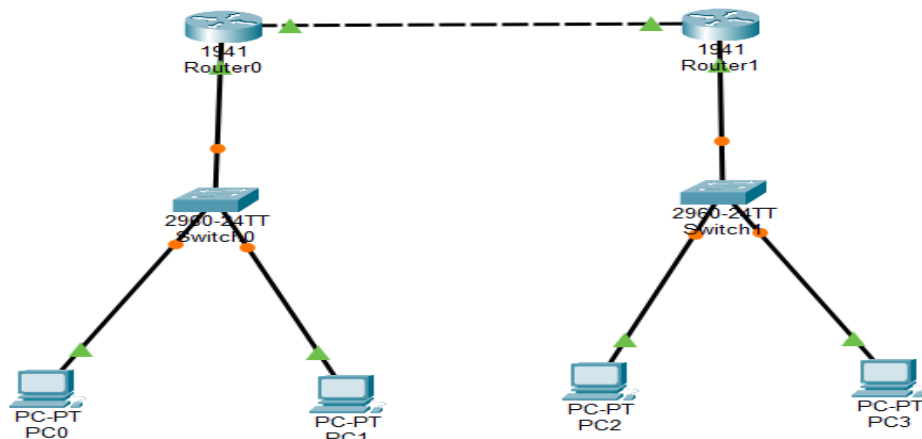
Router 0

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#
```

Router 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#
```

Network Topology



Ping Output:

```
PC0
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=11ms TTL=126
Reply from 192.168.2.3: bytes=32 time=11ms TTL=126
Reply from 192.168.2.3: bytes=32 time=17ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 17ms, Average = 13ms
C:\>
```

BGP (Border Gateway Protocol)

CLI:

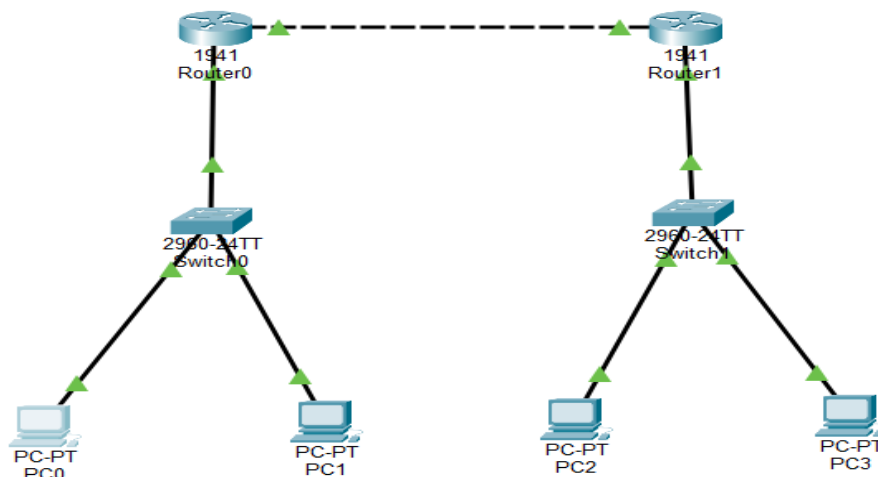
Router 0

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65001
Router(config-router)#neighbor 10.0.0.2 remote-as 65002
Router(config-router)#network 192.168.1.0 mask 255.255.255.0
Router(config-router)#
```

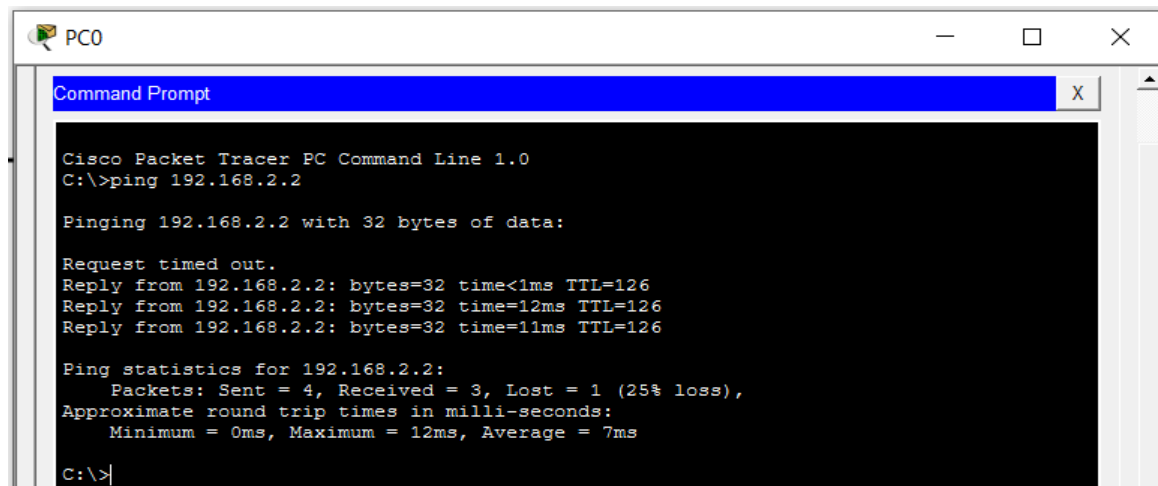
Router 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65002
Router(config-router)#neighbor 10.0.0.1 remote-as 65001
Router(config-router)#network 192.168.2.0 mask 255.255.255.0
Router(config-router)#
```

Network Topology



Ping Output:



Results

The lab successfully demonstrated the configuration of four major dynamic routing protocols: RIP, EIGRP, OSPF, and BGP. Routers and PCs were assigned IP addresses across two main subnets (192.168.1.0/24 and 192.168.2.0/24) connected via a 10.0.0.0/8 network. Each protocol was individually configured on Router0 and Router1 using Cisco Packet Tracer CLI. End-to-end connectivity was verified using ping commands, and successful replies between PC subnets (e.g., from PC0 to 192.168.2.2 and PC1 to 192.168.2.3) confirmed that routing tables were updated dynamically.

Protocol	Type	Metric Used	Convergence Speed
RIP	Distance-Vector	Hop Count (Max 15)	Slow
EIGRP	Advanced Distance-Vector	Bandwidth & Delay	Fast
OSPF	Link-State	Cost (based on Bandwidth)	Fast
BGP	Path-Vector	Path Attributes (AS path, policies)	Scalable/Stable

Discussion

Dynamic routing proved more efficient than static routing because routers automatically discovered and shared network updates without manual configuration. RIP, while simple to set up, relies on the Bellman-Ford algorithm and full-table updates every 30 seconds, which makes it inefficient for larger networks. EIGRP improved efficiency with partial and triggered updates, reducing unnecessary network traffic. OSPF maintained a complete network topology using Link-State Advertisements (LSAs) and the Dijkstra algorithm, which is ideal for large enterprise networks using hierarchical area designs. BGP is designed for routing between autonomous systems, relying on reliable TCP connections and policy-based decisions, making it suitable for large-scale and complex inter-network environments.

Conclusion

The lab achieved its objectives of configuring and verifying RIP, EIGRP, OSPF, and BGP. Dynamic routing protocols allow automatic path selection and quick adaptation to network changes. The experiment demonstrated that protocol choice depends on network size and requirements: RIP works for small networks, EIGRP and OSPF are ideal for internal corporate networks due to fast convergence and efficiency, and BGP is the standard for global inter-autonomous system routing.